RAAYA (رعايا): DEPI GRADUATION PROJECT

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Document Overview

This document details the specific software requirements necessary to engineer the "Ra'aya" frontend application. It acts as the critical bridge between our market research (what the users actually want) and our technical execution (what our developers need to build).

Inside, you will find a thorough **Stakeholder Analysis** that categorizes our main user groups—patients, nurses, and admins—and maps out their specific technical needs based on our survey data. More importantly, it translates these insights into actionable **Agile User Stories**, giving our React developers clear, scenario-based goals. The document further provides a strict list of **Functional Requirements**, defining the exact features we must build, such as the dual booking engine, the verification badge system, and the digital logbook. Lastly, it establishes the **Non-Functional Requirements**, setting non-negotiable standards for performance, accessibility (specifically tailored for elderly users), responsive design, and system reliability. Ultimately, this file will serve as the definitive reference guide for our development team throughout the implementation phase.

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3. Requirements Gathering & Analysis

This section defines the core needs, interactions, and constraints of the "Ra'aya" frontend platform. Following a strict **Data-Driven Design (DDD)** approach, these requirements were directly formulated based on a comprehensive UX/UI market survey conducted by our team, which gathered 155 responses from our target demographics.

3.1 Stakeholder Analysis

To build a highly tailored frontend experience, we identified our key stakeholders and analyzed their specific pain points and needs based on the survey data:

3.1.1 Patients & Patients' Families (77.4% of Respondents):

- **Core Needs:** Absolute trust, medical reliability, and ease of use.
- **Data Insight:** 71.7% expressed deep anxiety regarding the medical competence of unknown nurses, and 67.5% feared for their personal home safety. Furthermore, 79% requested a simplified, senior-friendly interface.
- **Frontend Implication:** The UI must prioritize "Trust Markers" (e.g., Verified Badges) above all other elements. The design system must be highly accessible, avoiding complex dropdowns in favor of large, readable typography and intuitive buttons.

3.1.2 Registered Nurses & Caregivers (14.8% of Respondents):

- **Core Needs:** Professional dignity, financial security, physical safety, and flexible scheduling.
- **Data Insight:** 47.8% suffer from a lack of official platforms that guarantee their rights. 95.6% demanded an emergency/SOS feature during home visits, and 82.6% preferred a Dark Mode interface for night shifts.
- **Frontend Implication:** The Nurse Dashboard must be strictly Mobile-First, featuring a built-in Theme Provider (Dark/Light mode), visual earnings tracking, and a highly accessible, swipe-to-confirm SOS UI component.

3.1.3 Platform Administrators & Medical Supervisors (7.7% of Respondents):

- **Core Needs:** Crisis management, efficient moderation, and financial transparency.
- **Data Insight:** 75% prioritize seeing urgent complaints immediately upon login, and 100% require an automated digital wallet system for weekly nurse payouts.
- **Frontend Implication:** The Admin Dashboard must be an "Exception-Based" UI, where critical alerts (negative reviews/SOS triggers) bypass standard data tables and appear at the very top as high-priority alert cards.

3.2 Agile User Stories & Use Cases

To guide the React development sprints, stakeholder needs have been translated into actionable Agile User Stories:

3.2.1 Patient Portal Scenarios:

- *As an elderly patient's relative*, I want to see a clear "Verified Badge" and read past reviews on the nurse's profile before confirming, *so that* I can feel 100% safe admitting them into my home.
- *As a user in a medical crisis*, I want to click an "Urgent Request" button (requested by 96.7%), *so that* I can bypass standard filters and get immediate medical assistance.
- *As a satisfied user*, I want a "Re-book" button next to my past appointments (requested by 95.8%), *so that* I can easily hire the same trusted nurse without repeating the search process.
- *As a skeptical user*, I want "Cash after session" to be the default payment option (preferred by 57.5%), *so that* I don't feel exposed to pricing exploitation.

3.2.2 Nurse Dashboard Scenarios:

- *As a nurse on a home visit*, I want an accessible "SOS Button" (requested by 95.6%), *so that* I can instantly alert the admin and the patient's family if the patient's health suddenly deteriorates.
- *As a freelance nurse*, I want to manage my shifts using an interactive "Calendar View" (preferred by 74%), *so that* I can visually organize my available working hours.
- *As a nurse working a 12-hour night shift*, I want a "Dark Mode" toggle (preferred by 82.6%), *so that* I can reduce eye strain while recording patient vitals.

3.2.3 Admin Panel Scenarios:

- *As an Admin*, I want negative reviews to appear at the top of my screen as red "Alert Cards", *so that* I can intervene and resolve patient complaints immediately.
- *As an Admin*, I want a filterable "Digital Wallet" data table, *so that* I can accurately track and manage the nurses' weekly financial settlements.

3.3 Functional Requirements

Based on the extracted User Stories, the React frontend must implement the following specific technical features:

1. **Dual Booking Engine:** The system must support two distinct booking flows: a standard multi-step scheduling flow (with specialty/gender filters) and an accelerated "Urgent/Emergency" booking flow.
2. **Smart Re-booking Automation:** A feature that automatically reads previous session data from the global state and populates the checkout form for rapid re-booking.
3. **The Digital Logbook Component:** Real-time data entry forms for nurses to input vitals (Blood Pressure, Glucose, Pulse), paired with interactive charts (e.g., Recharts) on the patient's side to visualize health trends.
4. **Verification UI System:** Reusable React Modals and Badges that explicitly display nurse credentials, syndicate licenses, and aggregated star ratings.
5. **Nurse Calendar & SOS Component:** Integration of a calendar library (e.g., react-big-calendar) for shift management, and a Swipe-to-Confirm SOS Floating Action Button (FAB) for emergencies.
6. **Role-Based Dashboards:** Distinct routing and interfaces for Patients (Vitals tracking), Nurses (Earnings & Calendar), and Admins (Alerts & Financial Tables).

3.4 Non-Functional Requirements

These constraints define the quality, performance, and usability standards of the frontend application, ensuring it meets the established KPIs:

1. **Usability & Accessibility (Senior-Friendly):** Given the target demographic, the UI must strictly adhere to WCAG accessibility guidelines. This includes high-contrast color schemes, large typography (text-lg or text-xl defaults for primary content), and large touch targets for mobile interfaces.
2. **Performance (Emergency Speed):** The application must have a fast First Contentful Paint (FCP < 2.0s). The "Urgent Request" and "SOS" features must be highly optimized to load instantly without heavy API blocking.
3. **Responsive Design (Mobile-First):** 100% responsiveness across all breakpoints. The Nurse Dashboard must be strictly designed "Mobile-First," as nurses will interact with the system entirely via smartphones during home visits.
4. **Reliability & State Consistency:** The application must utilize a robust global state management architecture (e.g., Redux Toolkit) to ensure that vital signs inputted by the nurse are reliably and instantly reflected in the patient's and doctor's dashboards without UI freezing or data loss.